55_Eritrea

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1 How has female employment in agriculture evolved relative to male employment in agriculture in Eritrea between 1991 and 2023?

1.1 Abstract

Using World Bank World Development Indicators (WDI) data, this study examines the evolution of employment in agriculture in Eritrea between 1991 and 2023, focusing on female versus male employment shares. These indicators provide insight into gender dynamics, structural shifts in the labor market, and the changing role of agriculture in the Eritrean economy. Over the period, both female and male employment in agriculture declined significantly, reflecting broader economic transformation, diversification of labor opportunities, and reduced reliance on subsistence agriculture. Female employment started slightly higher than male employment, and over time, male employment declined slightly faster, causing the gap between female and male shares to widen moderately. By 2023, female employment remained moderately higher than male employment, while the overall level of agricultural employment had fallen substantially. The trajectory illustrates both the decline of agriculture as a dominant employer and persistent gender differences in rural labor participation. These patterns offer a clear lens into Eritrea's labor market evolution, highlighting how structural changes affect men and women differently in traditional sectors, and suggesting that targeted policies may be needed to ensure equitable labor opportunities as the economy continues to diversify.

1.2 1. Question

How has female employment in agriculture evolved relative to male employment in agriculture in Eritrea between 1991 and 2023?

- **Female employment proxy**: Employment in agriculture, female (% of female employment, modeled ILO estimate)
- **Male employment proxy**: Employment in agriculture, male (% of male employment, modeled ILO estimate)

1.3 2. Data

- **Source**: World Bank World Development Indicators (WDI)
- **Indicators**:
  - Employment in agriculture, female (% of female employment, modeled ILO estimate)
  - Employment in agriculture, male (% of male employment, modeled ILO estimate)
- **Coverage**: Eritrea, 1991–2023

- **Notes**: National-level data only

1.4 3. Method

1. Filtered the dataset for Eritrea and selected female and male employment in agriculture indicators.
2. **Extracted relevant columns**: Year, Indicator Name, and Value.
3. Pivoted the dataset to create a side-by-side chronological comparison of female and male agricultural employment.
4. Produced a dual-line time series plot to visualize trends, relative magnitudes, and the widening gap between female and male employment over time.

(Analysis is descriptive; no causal inference applied.)

1.5 4. Results

- **Female employment in agriculture (% of female employment)**: Declined steadily over the period, starting slightly higher than male employment and remaining higher throughout.
- **Male employment in agriculture (% of male employment)**: Also declined steadily, at a slightly faster pace than female employment.
- **Comparison**: At the beginning of the period, female employment was slightly above male employment. By 2023, female employment remained moderately higher than male employment, reflecting a widening gap despite overall declines.

(Figure 1. Eritrea: Female vs. Male Employment in Agriculture (% of total employment), 1991–2023)

(Table 1. Pivoted dataset summary)

1.6 5. Interpretation

- Female employment consistently exceeding male employment highlights the persistent reliance of women on agriculture as a livelihood in Eritrea.
- Faster declines in male employment suggest a modest gender shift, possibly due to men moving to non-agricultural sectors or urban migration.
- The overall decrease in both series reflects structural changes in Eritrea's labor market, including diversification away from agriculture and increasing participation in services or informal sectors.
- Widening gender gaps emphasize the need to monitor equity in labor opportunities and design policies that support both male and female workers in adapting to evolving economic structures.

1.7 6. Limitations

- National aggregates may mask regional differences or sector-specific employment dynamics.
- WDI ILO-modeled estimates for earlier years may contain some uncertainty.
- Descriptive analysis does not establish causal links between sectoral shifts and policy, demographic, or economic factors.

1.8 7. Next Steps / Extensions

- Examine correlations between agricultural employment declines and rural-urban migration, industrial growth, or educational attainment.
- Compare Eritrea's trends with other East African countries to contextualize regional labor market transformations.
- Investigate sectoral policies and gender-targeted interventions that could support equitable employment as agriculture declines in relative importance.
- Analyze post-2023 data to assess whether trends continue or if female-male employment shares converge further.

[1]:
```python
# How has female employment in agriculture evolved relative to male employment␣
↪in agriculture in Eritrea between 1991 and 2023?

import pandas as pd
import matplotlib.pyplot as plt
import os

# Folders
data_raw_folder = "data_raw/"
data_clean_folder = "data_clean/"
figures_folder = "figures/"

# Load CSV
filename = "social-protection-and-labor_eri_filtered.csv"  # Filtered dataset␣
↪with only relevant rows
df = pd.read_csv(os.path.join(data_raw_folder, filename))

# Keep only needed columns
df = df[["Year", "Indicator Name", "Value"]]

# Convert Year and Value to numeric, drop invalid rows
df["Year"] = pd.to_numeric(df["Year"], errors="coerce")
df["Value"] = pd.to_numeric(df["Value"], errors="coerce")
df = df.dropna(subset=["Year", "Value"])

# Pivot indicators into separate columns
df_pivot = df.pivot(index="Year", columns="Indicator Name", values="Value").
↪reset_index()
df_pivot = df_pivot.sort_values("Year")

print("Pivoted Eritrea dataset:")
display(df_pivot)

# Interpolate missing values for smooth plotting (optional)
df_plot = df_pivot.interpolate(method='linear')

# Plot the indicators
```

```
plt.figure(figsize=(10,6))
plt.plot(df_plot["Year"], df_plot["Employment in agriculture, female (% of␣
 ↪female employment) (modeled ILO estimate)"],
         marker='o', linestyle='-', label="Employment in agriculture, female (%␣
 ↪of female employment)")
plt.plot(df_plot["Year"], df_plot["Employment in agriculture, male (% of male␣
 ↪employment) (modeled ILO estimate)"],
         marker='o', linestyle='-', label="Employment in agriculture, male (%␣
 ↪of male employment)")

plt.title("Eritrea: Female vs Male employment in agriculture (%) (1991-2023)")
plt.xlabel("Year")
plt.ylabel("Percentage")
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.savefig(os.path.join(figures_folder,␣
 ↪"eritrea_female_vs_male_employment_in_agriculture.png"))
plt.show()

# Save cleaned CSV
df_pivot.to_csv(os.path.join(data_clean_folder,␣
 ↪"eritrea_female_vs_male_employment_in_agriculture"), index=False)
```

```
Pivoted Eritrea dataset:

Indicator Name  Year  \
0               1991
1               1992
2               1993
3               1994
4               1995
5               1996
6               1997
7               1998
8               1999
9               2000
10              2001
11              2002
12              2003
13              2004
14              2005
15              2006
16              2007
17              2008
18              2009
19              2010
20              2011
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21              2012
22              2013
23              2014
24              2015
25              2016
26              2017
27              2018
28              2019
29              2020
30              2021
31              2022
32              2023

Indicator Name  Employment in agriculture, female (% of female employment)␣
 ↪(modeled ILO estimate)  \
0                                                       71.456464
1                                                       71.860331
2                                                       71.415520
3                                                       70.642207
4                                                       70.605325
5                                                       70.015014
6                                                       69.174435
7                                                       69.649924
8                                                       69.868275
9                                                       70.334733
10                                                      69.095604
11                                                      68.613565
12                                                      68.979949
13                                                      68.517273
14                                                      67.967494
15                                                      67.849809
16                                                      67.619328
17                                                      68.527992
18                                                      67.890978
19                                                      68.061705
20                                                      66.932210
21                                                      66.560215
22                                                      67.419716
23                                                      65.497841
24                                                      67.074089
25                                                      66.095250
26                                                      66.897931
27                                                      65.499159
28                                                      64.888098
29                                                      64.924851
30                                                      65.035716
31                                                      64.342189
```
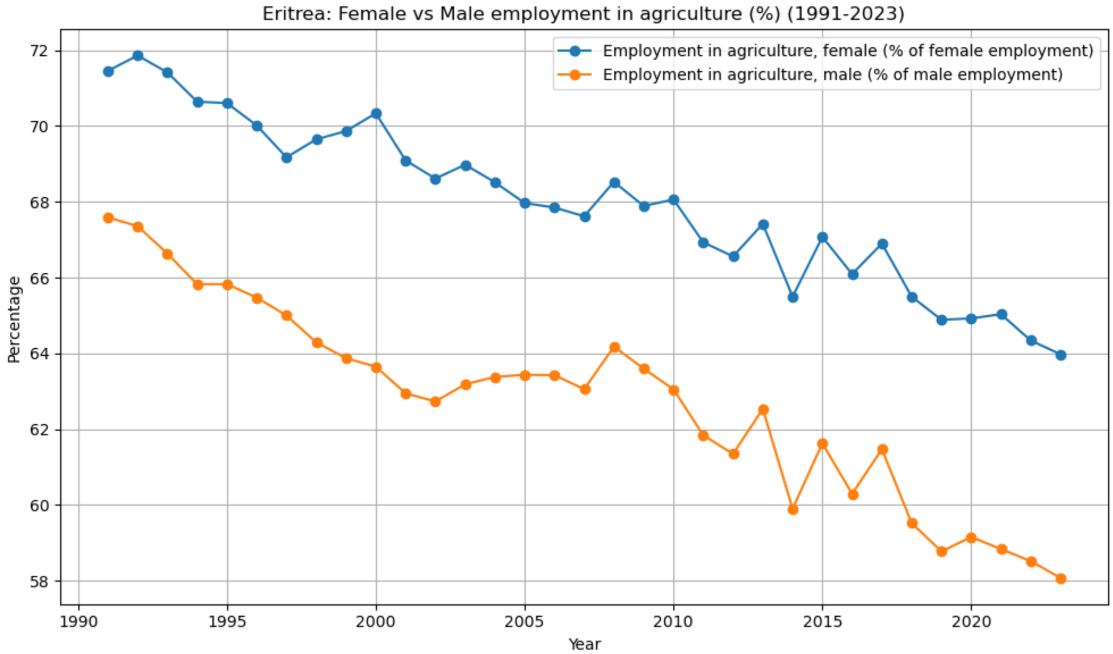

```
32                                                       63.977642

Indicator Name  Employment in agriculture, male (% of male employment) (modeled ILO estimate)
0                                                        67.592450
1                                                        67.356821
2                                                        66.634583
3                                                        65.826678
4                                                        65.828160
5                                                        65.474905
6                                                        65.000741
7                                                        64.292713
8                                                        63.876326
9                                                        63.651443
10                                                       62.949211
11                                                       62.732879
12                                                       63.185137
13                                                       63.379825
14                                                       63.437793
15                                                       63.427815
16                                                       63.056728
17                                                       64.178709
18                                                       63.598834
19                                                       63.051665
20                                                       61.842638
21                                                       61.347544
22                                                       62.539795
23                                                       59.900885
24                                                       61.628389
25                                                       60.297524
26                                                       61.473051
27                                                       59.526492
28                                                       58.775106
29                                                       59.156463
30                                                       58.839348
31                                                       58.518974
32                                                       58.077340
```


Eritrea: Female vs Male employment in agriculture (%) (1991-2023)
Employment in agriculture, female (% of female employment)
Employment in agriculture, male (% of male employment)
Percentage
Year
72
70
68
66
64
62
60
58
1990
1995
2000
2005
2010
2015
2020

[ ]: